

1 INTEGRATING FLIGHT AND CHARGING SCHEDULES IN URBAN AIR MOBILITY

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ABSTRACT

This paper investigates the simultaneous optimization of flight schedule and charging policy of electric vertical takeoff and landing aircraft for Urban Air Mobility operations. An optimization model is developed for a two-vertiport system to design an efficient flight schedule, including re-balancing flights, and a charging policy that serves a given demand with the objective of minimizing the required fleet size. The results highlight that charging aircraft to a lower State of Charge level during peak demand periods and to a higher level during off-peak times effectively reduces the fleet size, due to the non-linearity of battery charging times. The study includes a sensitivity analysis to determine the impact of various demand levels and flight times on the system. The results highlight the interplay between fluctuations in passenger demand and battery charging policies. Future work will integrate infrastructure constraints into the optimization model.

Keywords: Urban Air Mobility (UAM), Fleet Management, Charging Policy, Flight Schedule

1 INTRODUCTION

2 The aviation industry has been witnessing the transformative advent of Urban Air Mobility
 3 (UAM), catalyzed by the innovation of efficient and compact electric vertical take-off and landing
 4 (eVTOL) aircraft, earmarked as a promising solution for swift urban transportation (1, 2). While
 5 UAM's potential to mitigate surface traffic congestion and foster a sustainable urban environment is
 6 significant, its implementation is fraught with systemic and infrastructural challenges (3). Among
 7 these are the needs for effective fleet management, precise flight scheduling, and the development
 8 of robust charging infrastructure. The success of UAM further depends on the formulation of a
 9 comprehensive, high-dimensional charging policy, incorporating variables such as time, location,
 10 number of charging tasks, and charging intervals, which is crucial to the feasibility and operability
 11 of UAM systems (4).

12 This paper presents a comprehensive scheduling methodology that addresses existing re-
 13 search gaps in UAM operations. We propose a dynamic model that concurrently plans flight and
 14 charging schedules in a multi-vertiport system, employing a non-linear charging model to reflect
 15 realistic charging behaviors, providing an enhanced representation over traditional linear assump-
 16 tions. Demand scenarios are generated based on realistic airport access patterns. The numerical
 17 analysis focuses on a two-vertiport system. The main objective is to design a flight and charging
 18 schedule such that the fleet size is minimized while respecting constraints related to demand and
 19 infrastructure. The research makes significant contributions to the discourse on effective UAM
 20 operations, offering insights and solutions that address existing challenges and lay the groundwork
 21 for more sustainable, efficient UAM systems in urban settings.

22 The paper is organized as follows. In 'Literature Review', we review existing literature,
 23 identifying the gaps that our study addresses. The subsequent section 'Optimization Model' dis-
 24 cusses the dynamic equation that models the number of aircraft in different states and the decision
 25 variables, including flight scheduling and charging planning. In the section 'Numerical Analysis',
 26 we discuss the process of deriving the input parameters to the optimization model and delve into
 27 the implications of our results for UAM operations management. Finally, under 'Conclusion and
 28 Discussion' we summarize our key findings, discuss their implications for UAM system opera-
 29 tions, and suggest potential avenues for future research.

30 LITERATURE REVIEW

31 UAM represents a compelling advance in aviation technology. It leverages compact, eV-
 32 TOL aircraft to provide rapid urban transit. However, the realization of UAM's potential depends
 33 on the successful negotiation of numerous operational and infrastructural challenges (4). Essential
 34 considerations for efficiency encompass fleet management strategy, flight dispatching policy, and
 35 charging schedules.

36 Many insightful investigations have searched for optimal infrastructure parameters for UAM
 37 systems. Li et al. (5) test the feasibility of the UAM system under various demand profiles and an-
 38 alyze how operation parameters, such as fleet size and vertiport capacity, affect the overall system
 39 performance. Similarly, Husemann et al. (6) develop an optimization model that considers the total
 40 cost of operating UAM systems to design optimal infrastructure, such as the battery capacity of the
 41 aircraft and the power of the chargers. Instead of optimization, Swaid et al. (7) adopt a simulation
 42 approach to derive the minimum fleet size by assuming an unlimited pool of aircraft and charging
 43 power of 50 kW. On the tactical side, Wu and Zhang (8) develop a local charging scheduler at a
 44 vertiport that minimizes the total passenger waiting time.

Significant strides have been made in various aspects of designing UAM operations. Husemann et al. (9) apply Bipartite Matching Dispatcher to fleet assignment with online simulation. Song and Yeo (10) consider various approach control methods and their corresponding performances. Paul and Chowdhury (11) encode the vertiport network in graphs and use reinforcement learning to conduct fleet assignment, given that the charging and demand constraints are satisfied. Kim (12) further consider a heterogeneous fleet composition and designs a genetic algorithm to arrive at near-optimal scheduling policies. Additionally, ongoing efforts have also been concerned with arrival sequencing (13).

Although many have addressed the charging constraint and the demand imbalance of UAM systems, there lacks an integrated operation policy. Studies have overlooked the non-linear nature of charging power and neglected the charging scheduling in relation to flights. We aim to address these gaps by proposing a more comprehensive approach to UAM operations planning. We use a realistic non-linear charging model to derive charging policies in conjunction with flight scheduling that considers demand imbalance.

OPTIMIZATION MODEL

In this paper, we tackle an operation planning problem. Specifically, given a time-varying demand profile for the number of flights between two vertiports, we determine the number of eVTOLs and their flying schedules and charging policies while minimizing the number of aircraft and the number of flights. For this, an operating day is divided into a finite number of equal-length time steps and eVTOL flying and charging decisions are made at each time step. Our cost function seeks to minimize the fleet size required for a flight demand while also reducing the number of re-balancing flights without increasing the fleet size. We assume a non-linear aircraft charging and energy consumption model and embed the two models as constraints in the optimization. As eVTOLs are electric-powered, we also consider a finite set of States of Charge (SoCs) for eVTOLs. To formulate the problem in an optimization model, the following notations are introduced in table 1.

The objective of the optimization model is to find a finite horizon fleet dispatch and charging policy that minimizes the number of eVTOL aircraft and the number of flights while being constrained by the flight demand, passenger demand, reserve State of Charge (SoC) requirement, and cyclo-stationarity, i.e., we require the state at the beginning and the end of the optimization time period to be the same so that the optimal policies can be repeated.

$$\min \sum_i \sum_k n_i^k(t=0) + \sum_i \sum_x \sum_y C_i^{xy}(t=0) + \alpha \cdot \sum_t \sum_i \sum_j \sum_k u_{ij}^k(t) \quad (1)$$

The first two summation terms in the objective function represent the total number of eVTOL aircraft, based on time step $t = 0$, i.e., at the beginning of an operating day. Because no aircraft is flying at $t = 0$, the sum of aircraft idling and charging at $t = 0$ constitutes the fleet size. The third term in the objective function represents the total number of flights with a small weight α , which has the purpose of constraining the solution space so that between two solutions that yield the same fleet size, the solution that minimizes the number of flights is preferred. We set $\alpha = 0.00001$ because α has to be small enough such that $\alpha \cdot \text{number of flight}$ does not exceed 1.

TABLE 1: Notations

	Notation	Explanation
Decision Variables	$C_i^{xy}(t)$	Number of aircraft at vertiport i that begin to charge at time step t with an initial SoC of x and a target SoC of y
	$u_{ij}^k(t)$	Number of aircraft departing for vertiport j from vertiport i at time step t with SoC k
Parameters	$n_i^k(t)$	Number of idle aircraft at vertiport i with SoC k at time step t
	$f_{ij}(t)$	Number of flights departing from vertiport i to vertiport j at time step t required to satisfy travel demand
	γ_k	Charging time needed to transition from SoC_{k-1} to SoC_k
	K	Number of SoCs after discretization
	τ_{ij}	Flight time from vertiport i to j in time steps
	κ_{ij}	Reduction in SoC for a flight from vertiport i to j
	T	Planning horizon $T = \text{number of time steps} + \max_{ij} \tau_{ij} + 1$

Minimization of the objective function is subject to the following constraints:

$$n_i^k(t) = n_i^k(t-1) + \sum_{j \in V - \{i\}} u_{j,i}^{k+\kappa_{ji}}(t - \tau_{ji}) - \sum_{j \in V - \{i\}} u_{ij}^k(t) + \sum_{x=0}^{k-1} C_i^{xk}(t) - \sum_{i=x+1}^k \gamma_i - \sum_{y=k+1}^K C_i^{ky}(t), \quad \forall i, t \quad (\text{Conservation of Aircraft}) \quad (2)$$

$$\sum_{k \in \{1, \dots, K\}} u_{ij}^k(t) \geq f_{ij}(t), \quad \forall i, j, t \quad (\text{Passenger Demand Constraint}) \quad (3)$$

$$u_{ij}^0(t) = 0, \quad \forall i, j, t \quad (\text{Reserve SoC Constraint}) \quad (4)$$

$$n_i^k(0) = n_i^k(T), \quad u_{ij}^k(0) = u_{ij}^k(T), \quad C_i^{xy}(0) = C_i^{xy}(T), \quad \forall i, j, k, x, y \quad (\text{Cyclo-stationarity}) \quad (5)$$

The dynamic equation that governs the evolution of the system is given by constraint (2), which models the state of the aircraft in relation to the charging policy and the flight schedule. We model the number of aircraft that enter a state of idle at each vertiport at each time step of a certain SoC as a combination of the following terms: (A) the number of idling aircraft that are carried over from the previous time step. (B) the number of aircraft that will arrive at vertiport i at time step t of SoC k . This comes from the flight schedule. (C) the number of aircraft that depart from vertiport i of SoC k . (D) the number of aircraft that complete charging to SoC k at time step t at

vertiport i . (E) the number of aircraft that are committed to charging at vertiport i starting from SoC k . Constraint (3) ensures that flight demand for each vertiport pair (i, j) , measured as the required number of flights $f_{i,j}(t)$, is satisfied at each time step. Constraint (4) ensures that aircraft cannot fly when the SoC level equals 0, which is the reserve SoC. Constraint (5) ensures that the number of aircraft in different states and SoCs are the same at the beginning and the end of the time horizon. These constraints are essential to ensure the feasibility and robustness of the flight schedule and charging policy.

In practice, the above optimization may be performed at the beginning of an operation day or earlier before the actual operation. The model can be extended to include constraints that reflect the physical and operational limits of the vertiport network and the charging infrastructure. For instance, a constraint on the maximum number of charging pads at each vertiport can influence the distribution of charging activities across the vertiports. Similarly, a constraint on the maximum number of take-offs and landings at a time step at a vertiport can influence the distribution of scheduled flights. Exploring these additional constraints is part of our ongoing research.

NUMERICAL ANALYSIS

We demonstrate the implementation of the optimization model with a two-vertiport system in the context of a large metropolitan in the US. For confidentiality, the actual names have been replaced by surrogates. One vertiport is located near the main airport (APT) of the city. The other vertiport is located in the Central Business District (CBD) of the city. The UAM flight demand is derived from the arrival and departure schedule at the airport on a busy day in 2019. We discretize the 24 hours in the day into 288 5-minute steps in the optimization model. The following section details the process of deriving the input parameters to the model, including the flight demand, the charging model, and the flight profile.

Input Models

Demand Shaping

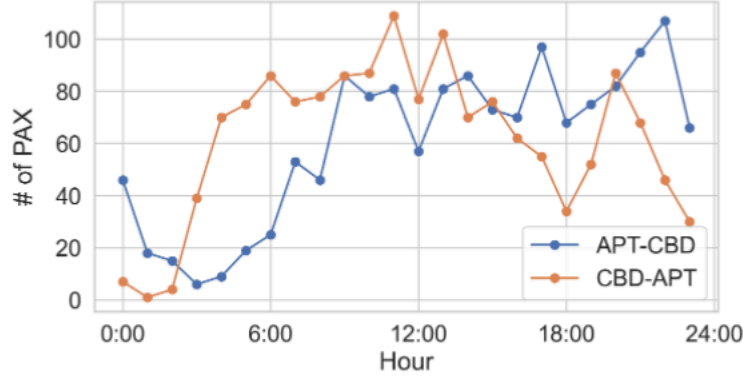
In this two-vertiport system, UAM demand is derived from airline passengers departing from and arriving at APT. Arrivals result in demand from APT to CBD and departures create demand from CBD to APT. With the departure and arrival flight schedule of a given airport and the estimated total UAM demand, we can derive the density distribution of UAM passenger demand during the day. Assuming that each airline flight has a certain number of passengers taking UAM (varying based on the seating capacity of the aircraft), we generate the number of passengers from each airline flight that takes UAM. Then, we sample the lead or lag time for those passengers from skewed normal distributions. For UAM passengers departing from CBD for APT, we define their arrival time at the airport vertiport as the departure time of the airline flight from the airport minus the lead time. For UAM passengers arriving at the airport and going to CBD, we define their arrival time at the vertiport as the time of the airline flight arriving at the gate plus the lag time. Algorithm 1 shows the process of generating passenger arrival times at the vertiport. The expected number of passengers for aircraft type i can be calculated as follows:

$$\lambda_i = \frac{\text{number of seats on aircraft type } i}{\text{total number of arrival/departure seats}} \cdot \text{estimated directional UAM demand} \quad (6)$$

The 2013 Airport Cooperative Research Program (ACRP) report is used to estimate the lead and lag time distributions (14). The lead time follows a skewed normal distribution with $\mu = 93$,

Algorithm 1 Passengers' Arrival at the Two Vertiports

-
- 1: **for** every airline flight arriving/departing at an airport **do**
 - 2: Determine number of passengers taking UAM from $Poission(\lambda_i)$
 - 3: Generate the lag/lead time for each passenger from the skewed normal distributions
 - 4: Add lag/lead time to the time of arrival/departure
 - 5: **end for**
-

**FIGURE 1:** Passenger Arrivals at the Airport Vertiport

1 $\sigma = 40$, and the skewness parameter $\alpha = 3$, which results in a median lead time of 120 minutes
2 (14). The ACRP report does not provide a distribution for lag time. By taking the provided mean
3 lag time, 21 minutes, and assuming a transfer time of 10 minutes from the curb of the airport to
4 the vertiport, we arrive at a skewed normal distribution with $\mu = 31$, $\sigma = 2.12$, and $\alpha = 3$ (14).
5 Figure 1 shows the number of passengers arriving at both vertiports on an hourly basis
6 assuming that the directional demand is 1,500 passengers as a baseline suggested by Preis and
7 Hornung (15). We then apply a deterministic dispatch policy where a flight is dispatched when the
8 eVTOL is full or when the first passenger has waited for more than a time threshold. Assuming an
9 eVTOL capacity of four passengers and a threshold waiting time of 5 minutes, Figure 2 shows the
10 distribution of flight demand in both directions throughout the day on an hourly basis.

Charging Model

12 We discretize battery charging levels into discrete States of Charge (SoC) and model the
13 charging time required to reach a given SoC from a lower SoC. We place particular emphasis on
14 the non-linearity of these charging times.

15 To model the charging process of eVTOLs, we use the battery on an electric vehicle (EV).
16 Currently, Lucid Air Dream Edition has the largest battery capacity and is the fastest-charging
17 electric vehicle (EV) available. It has a 118 kWh battery and requires 33 minutes to get charged
18 from 10% SoC to 80% SoC with a 350 kW charger (16).

19 The charging times of urban air vehicles are not published in the open literature. Using
20 experimental data (16), we simplify the SoC - charge rate relation as given in equation 5.1.2. In
21 this model, we assume that the charger's charge rate will be constant until 20% SoC, then the
22 charge rate decreases linearly until the battery is fully charged. Based on this model, we get the

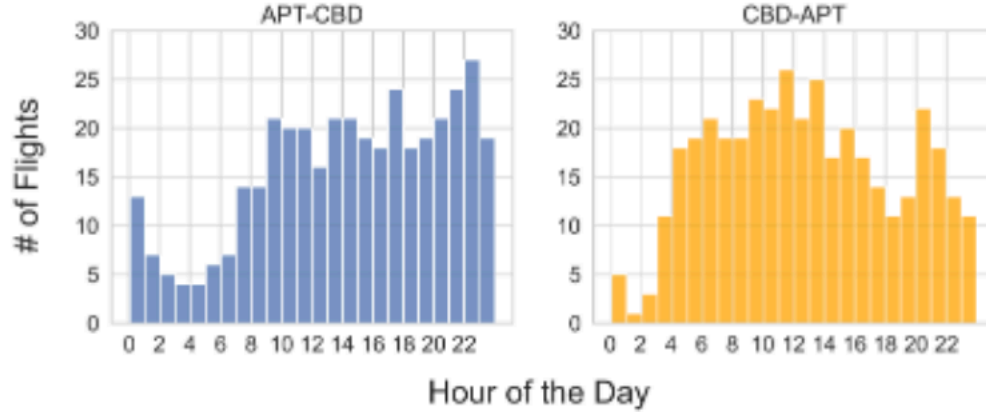


FIGURE 2: Hourly Flight Demand with 5-Minute Waiting Time Threshold

1 charge rate from SoC by using a piece-wise linear function.

$$f(x) = \begin{cases} M & \text{if } x \leq 20\% \text{ SoC} \\ M - S * (x - 20) & \text{if } x > 20\% \text{ SoC} \end{cases} \quad (7)$$

2 where M is the maximum charge rate and S is the slope of the line.

3 The substantial energy demands of eVTOLs and proposed short turnaround times necessi-
 4 tate the use of high-capacity chargers (17). 350 kW chargers are the highest charge rate chargers
 5 that are put into the service. Therefore, in this paper, we assume 350 kW chargers for eVTOL
 6 charging.

7 Using a 350 kW charge rate and integrating the charging power curve for a 160 kWh battery,
 8 we derive the cumulative charging time as given in figure 3. We then formulate the charging time
 9 vector, γ_k , by designating 32 levels of non-reserve SoCs above a 20% reserve requirement.

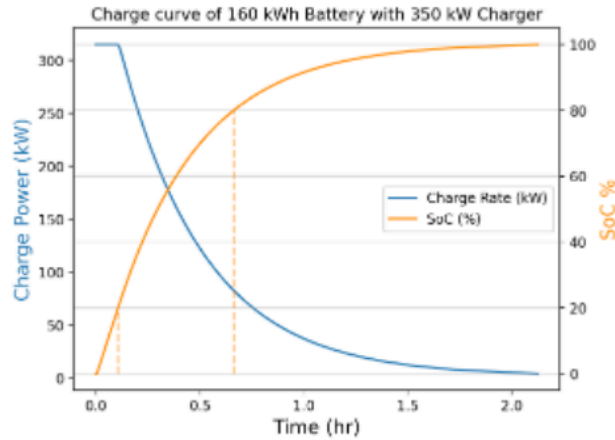


FIGURE 3: Time vs SoC curve

10 *Flight Profile*

11 In our study, we utilize a model-driven approach to evaluate the energy consumption of the
 12 aircraft, using SoC as an indicator of energy usage. This approach defines one SoC as 2.5% of the

total battery capacity, allowing for a detailed representation of energy consumption. Such granularity enhances the precision in operational planning and decision-making processes, as it provides a more nuanced understanding of the energy dynamics within the aircraft. Our computation of the SoC consumption with respect to different flight distances is built upon the energy model introduced by Sripad and Viswanathan (17) for the Joby 5-seater aircraft. Furthermore, we incorporate specific flight parameters for the same aircraft in our model, such as maximum power speed for ascent and descent phases, and maximum range speed for cruising. These parameters enable us to accurately estimate the flight time over various distances, thereby contributing to a comprehensive energy consumption analysis.

In the subsequent phase of our study, we calculate the flight time for distances ranging from 12 to 108 miles, incremented in steps of 12 miles. This range corresponds to 2 to 9 time steps, each with its associated SoC consumption as shown in Table 2.

Baseline Scenario Results

The baseline scenario considers the distance of a flight to be 12 miles based on the actual locations of the two vertiports considered. It takes an eVTOL 10 minutes (two time steps) to make a trip, with an energy consumption of 10% of the battery capacity (equivalent to 4 SoCs). We further assume that the directional demand is 1,500 passengers to be an illustrating nominal, which is used to generate the flight demand using the process described previously. Table 2 lists the parameters used in the numerical analysis.

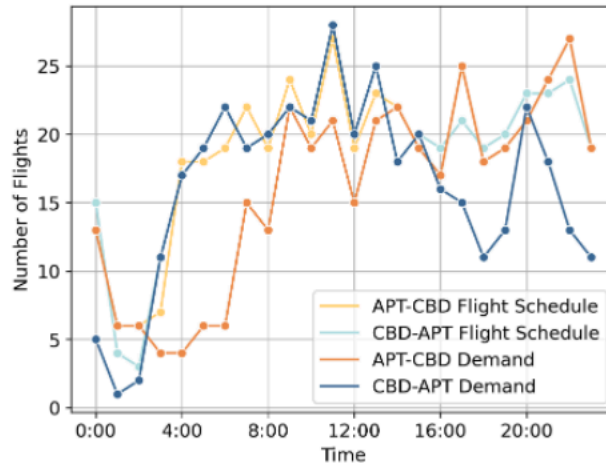


FIGURE 4: Optimal Flight Schedule with Re-Balancing

Figure 4 shows the number of flights from the flight schedule on an hourly basis. It can be observed that the solution that the optimization model produces set the flight direction that has a higher demand as the upper bound for the number of flights and creates re-balancing flights in the other direction. Therefore, re-balancing flights from the airport to the CBD only take place in the early morning as more passengers are traveling to the airport. On the other hand, we see more re-balancing flights from the CBD to the airport in the late afternoon and early morning as the airline passenger arrivals at the airport increasingly outnumber passenger departures.

The optimization model outputs a 5-dimensional charging policy, dictating the number of aircraft to be charged from an initial SoC to a target SoC at a certain vertiport and time step. Figure

TABLE 2: Input Parameters

	Directional Demand (PAX)	Energy Consumption (% Full Charge)	Flight Time (min)	Flight Distance (mile)
Baseline	1,500	10	10	12
Demand Sensitivity (DS) 1	500	10	10	12
DS2	1,000	10	10	12
DS3	2,000	10	10	12
DS4	2,500	10	10	12
Flight Time Sensitivity (FTS) 1	1,500	15	15	36
FTS2	1,500	27.5	20	48
FTS3	1,500	32.5	25	60
FTS4	1,500	37.5	30	72
FTS5	1,500	42.5	35	84
FTS6	1,500	45	40	96
FTS7	1,500	52.5	45	108

5 illustrates the charging events that occur over a 24-hr period, aggregated over the 2 vertiports and every 4 hours. It reveals that aircraft are charged to a higher SoC during the early morning when passenger demand is lower. On the other hand, aircraft are charged to a lower SoC during the peak demand periods, particularly between 12:00 pm and 4:00 pm.

Figure 6 presents an overview of the quantity of aircraft engaged in charging operations throughout the day, alongside the corresponding time durations, measured in minutes, that these charging events entail. In the early morning, when demand is comparatively modest, aircraft are charged to more elevated SoCs. This process demands a more extended charging period, a consequence of the charging rate's non-linear behavior. As demand escalates throughout the day, with a pronounced peak particularly between 12:00 pm and 4:00 pm, a significantly larger number of aircraft are charged to a relatively low SoC. The devised solution optimizes the number of aircraft required by minimizing the charging time during periods of peak demand. Concurrently, the solution capitalizes on off-peak times by charging available aircraft to higher SoC levels. This strategy offers a balanced approach, effectively addressing the dynamic demands of eVTOL operations.

Combining the two decision variables, $u_{ij}^k(t)$ (flight schedule) and $C_i^{xy}(t)$ (charging policy), we compute the state of aircraft in the system to derive the fleet size and the infrastructure require-

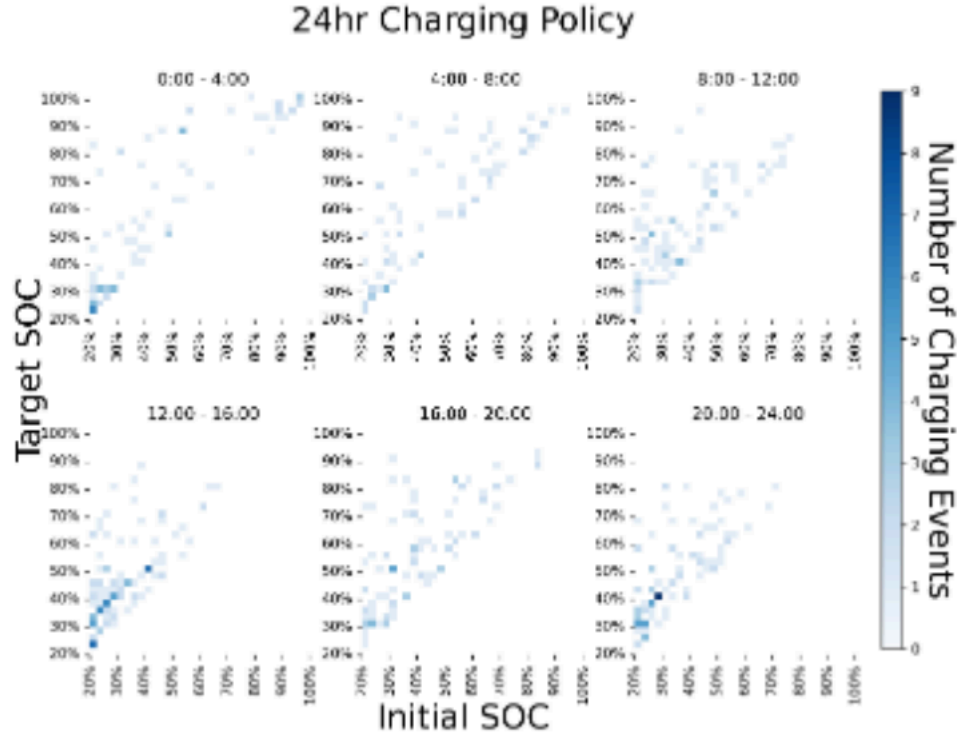


FIGURE 5: Charging Policy with 1,500 passenger Demand

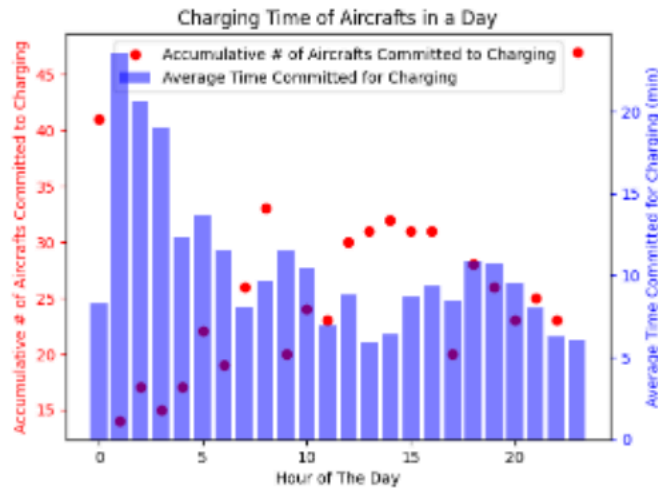


FIGURE 6: Charging Demand with 1,500 PAX Demand

ment. Given the flight time, which is 2 time steps, and the charging model described previously, we can calculate the number of aircraft in each state at a given time step. From figure 7, we learn that the optimal fleet size is 12. Additionally, the number of parking/charging pads required at the airport vertiport and the CBD vertiport are 6 and 12, respectively, which are the maximum number of aircraft on the ground at any given time at the two vertiports. These aircraft on the ground consist of those that are idling and are being charged. The number of aircraft in flight is the number of aircraft that left the vertiport but have not reached their destination. Note that the optimization

1 model does not consider infrastructure constraints. Integrating infrastructure constraints into the
 2 program will be a focus for the next phase of our work.

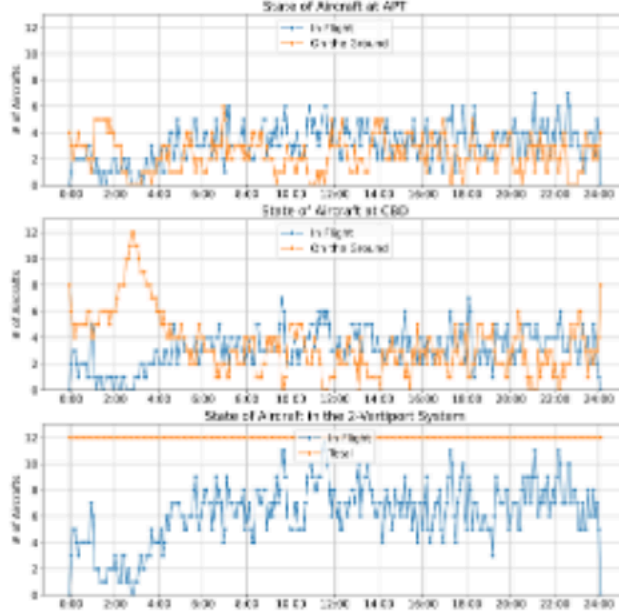


FIGURE 7: State of Aircraft at Vertiports and in the System

3 Comparison with alternative strategies for flight scheduling and aircraft charging

4 To further understand the benefit of the simultaneous optimization of flight scheduling and
 5 aircraft charging through our optimization model, this section compares the model results with
 6 results under alternative strategies for flight scheduling and aircraft charging. Specifically, four
 7 alternative strategies are considered:

- 8 • Strategy 1 does not use re-balancing flights to resolve the demand imbalance, and charges
 9 each aircraft to full after each flight. To implement this strategy, we add a constraint
 10 such that only aircraft of full SoC can depart: $u_{ij}^k(t) = 0, k \in \{0, 1, 2, \dots, K-1\}$ and
 11 $\sum_k u_{ij}^k(t) = f_{ij}(t) \quad \forall i, j$.
- 12 • Strategy 2 is a modification of the first. While it still does not consider re-balancing
 13 flights, strategy 2 dictates that aircraft are only charged and charged to full when their
 14 SoC falls to the reserve capacity. This strategy introduces an element of energy conser-
 15 vation, potentially extending battery lifespan by circumventing unnecessary charging cy-
 16 cles. We add a constraint on the charging policy such that only charging from the reserve
 17 SoC to the full SoC is allowed: $C_i^{xy}(t) = 0, \forall x \neq 0, y \neq K$ and $\sum_k u_{ij}^k(t) = f_{ij}(t) \quad \forall i, j$.
- 18 • Strategy 3 uses the optimized flight schedule from the optimization model but charges
 19 aircraft to full after each flight. In this case, we present an optimization formulation that
 20 schedules flights without consideration for charging by giving the following constraint to
 21 the proposed optimization model: $C_i^{xy}(t) = 0$ for $y \neq K$
- 22 • Strategy 4 uses the optimized flight schedule from the optimization model and charges
 23 aircraft to full when their SoC falls to the reserve capacity. This strategy is comprehen-
 24 sive, as it combines elements of all previous strategies to strike a balance between energy

usage, battery longevity, and operational efficiency. Still, the dimension of the charging policy is smaller than the optimization output, indicating more room for improvement: $C_i^{xy}(t) = 0 \forall x \neq 0, y \neq K$.

In Table 3, we present the outcomes of these strategies. Charging to full after each flight drastically increases the fleet size as it takes as much as eight times longer to transition from the 31st SoC to the 32nd than to transition from SoC 0 to SoC 1. Setting a threshold to initiate charging greatly reduces the fleet size as more charging occurs at low SoCs where charging rates are higher. Furthermore, implementing the optimized flight schedules decreases the fleet size by minimizing the total number of flights in the system, thus reducing energy consumption. The result from the comparative analysis highlights the importance for considering both an optimized flight schedule and an efficient charging policy. It also suggests that a simplified charging policy – fully charging from reserve SoC – is almost as efficient as the optimized policy.

TABLE 3: Benchmark Analysis

Strategy	Fleet Size	No. of Pads
Strategy 1: on-demand, full charge	116	189
Strategy 2: on-demand, full charge when $\leq 20\%$	82	134
Strategy 3: optimized schedule, full charge	83	127
Strategy 4: optimized schedule, full charge when $\leq 20\%$	13	22
Optimized schedule and charge (our model)	12	18

Sensitivity to Demand

Due to the limited capacity of the UAM aircraft that are currently in development, accommodating a larger demand leads to an increase in the number of flights. Hence, we are interested in understanding the impact of demand on the 2-vertiport system. We assume a directional demand of 500, 1,000, 2,000, and 2,500 passengers in addition to the 1,500 passengers used in the baseline scenario as inputs. We analyze the solutions to the program, specifically focusing on the fleet size, the number of pads needed, the ratio of revenue aircraft miles to total aircraft miles, and load factor. We defined load factor as the number of passenger per scheduled flight, including the re-balancing flights. Table 4 shows that as demand increases, the fleet size scales sublinearly while the utilization remains around 85%. The average load factor also increases as we increase the demand. The results show that in a two-vertiport system, as the demand increases, we obtain economy of scale. This is a result of the dispatching policy to release a flight within five minutes of a passenger arrival, which results in more empty seats when demand is lower. However, the average load factor stabilizes at around 3.4, given the set of input parameters we use.

Sensitivity to Flight Distance

We analyze the sensitivity of our results to flight distance, which is captured as flight time in our model, for a fixed demand of 1,500 passengers in each direction. A total of eight flight time sensitivity (FTS) scenarios are examined, with energy consumption ranging from 4 to 21 SoC steps

TABLE 4: Demand Sensitivity Analysis

One-Way Demand	Fleet Size	Number of Pads	Ratio of Revenue Aircraft Miles to Aircraft Miles	Average Load Factor
500	5	5	85.4%	2.39
1,000	9	13	85.2%	3.03
1,500	12	18	85.3%	3.32
2,000	15	22	83.8%	3.32
2,500	19	29	84.9%	3.42

TABLE 5: Flight Time Sensitivity Analysis

Distance (mi)	Fleet Size	Number of Parking/Charging Pads	Ratio of Revenue Aircraft Miles to Aircraft Miles
12	12	18	85.3%
36	19	29	88.6%
48	26	38	89.2%
60	32	44	89.9%
72	38	51	88.6%
84	43	56	90.9%
96	48	61	91.3%
108	60	76	90.3%

1 and flight times from 2 to 9 time steps. The specifics of these scenarios, including their SoC and
2 time step values, are shown in Table 2.

3 We explore the relationship between flight time and the fleet size and charging pad param-
4 eters required for operations. Additionally, we analyze a scenario of battery swap, where we assume
5 charging completes instantaneously and a case of linear charging times. In Table 5, we present the
6 results of the flight time sensitivity analysis. The tabular representation highlights trends in fleet
7 size and the number of pads as flight distance rises.

8 Commencing with the baseline scenario in which the flight distance is 12 statute miles, the
9 fleet size is 12, and the number of pads is 18. As flight distance increases to 36 miles (FTS 1), fleet
10 size and the number of pads increase to 19 and 29, respectively. However, a noteworthy change

occurs as the flight distance stretches to 48 miles (FTS 2). The rate of increment in fleet size and the number of pads becomes distinctly more pronounced, with the fleet size inflating to 26 and the number of pads reaching 38. This amplified rate of expansion is sustained when the flight distance rises to 60 miles (FTS 3), causing the fleet size and the number of pads to ascend to 32 and 44, respectively. The sensitivity analysis shown in Table 5 verifies a clear pattern of growth in fleet size, and the number of pads parallels an increase in flight distance.

Notably, the efficiency of aircraft usage, as measured by the ratio of revenue aircraft miles to total aircraft miles, also varies with flight time, generally increasing as flight time increases. This ratio, which reflects the proportion of total flight distance that is devoted to revenue-generating flights, ranges from 85.3% to 91.4%. This trend shows that longer flights induce more efficient use of aircraft, due to the greater cost of repositioning flights.

The convex relation between flight distance and resource requirements reflects the non-linear charging relationships, as shown in Figure 8. On the other hand, under both battery swap and linear charging time assumptions, the fleet size increases proportionally to distance. The juxtaposition of these disparate growth patterns accentuates the significance of non-linear charging times in driving fleet requirements.

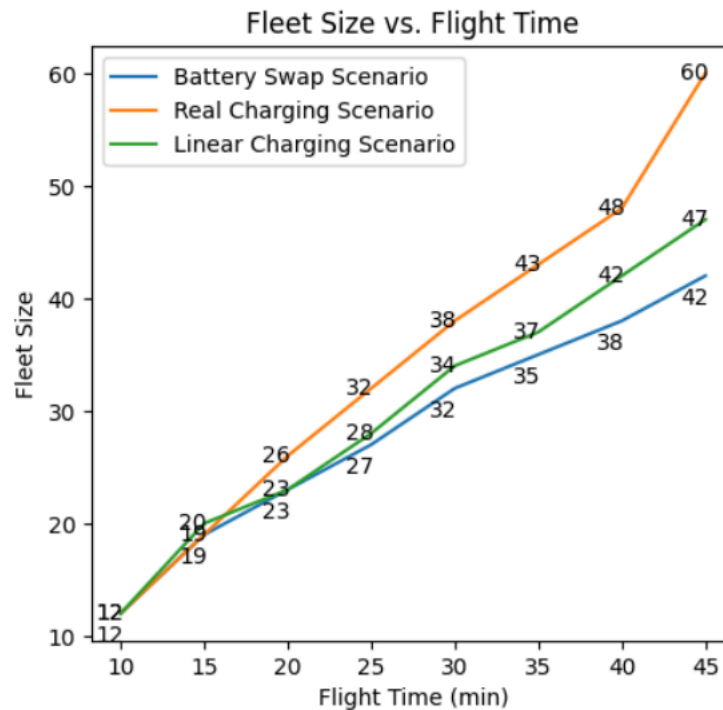


FIGURE 8: Real Charging Scenario vs Battery Swap Scenario

CONCLUSION AND DISCUSSION

This study presents a novel approach to tackling the challenges of flight and charging scheduling in the rapidly evolving field of Urban Air Mobility (UAM). The proposed optimization model for a two-vertiport system has been shown to improve operational efficiency by minimizing the number of required aircraft and flights. Our results, derived from a scenario reflective of a busy commercial airport in the U.S., highlight the model's applicability in a real-world setting.

1 The model's strength lies in its integration of both flight and high-dimensional charging
2 schedules, a feature that is often overlooked in conventional models. This integration provides
3 more efficient operation strategies for UAM systems as it minimizes both the fleet size and the
4 number of flights with considerations for demand imbalance. These features are particularly im-
5 portant when charging times are non-linear.

6 The findings of this study open up several avenues for future research. First, while the
7 model is tested on a single two-vertiport system, further testing and validation could be carried
8 out in more complex multi-vertiport systems with varying demand patterns and constraints. This
9 would help to ascertain the model's scalability and adaptability to diverse urban settings. Second,
10 the model could be enhanced by incorporating more real-world considerations such as weather
11 conditions, air traffic controls, and potential technical failures. Such improvements would make the
12 model more robust and practical for practical UAM operations. Third, the economic implications
13 of implementing such an optimization model in UAM operations could be explored. This would
14 involve an analysis of the trade-offs between fleet size, repositioning flights, pad requirements,
15 passenger waiting time, and other factors.

16 Within a broader context, this study provides valuable insights into the efficient manage-
17 ment of UAM operations. The integration of flight and charging schedules as proposed in this study
18 could potentially lead to significant improvements in the operational efficiency and sustainability
19 of the UAM industry.

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